**SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical Sciences**
**Chennai- 602105**

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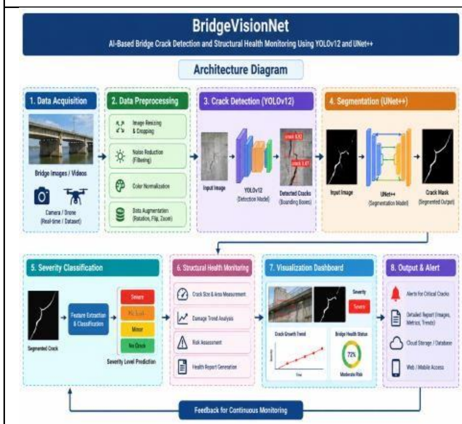
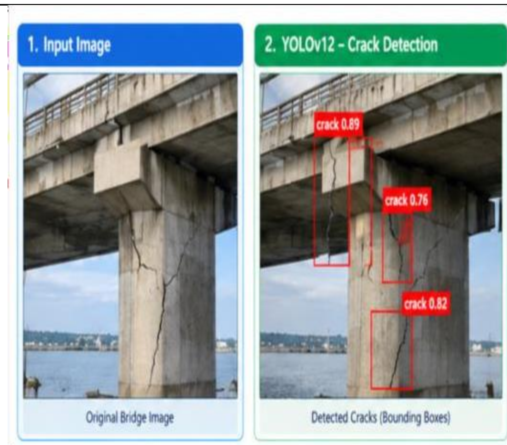
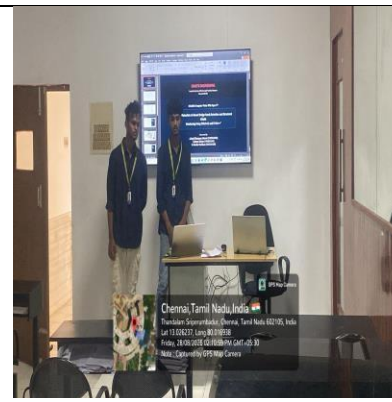
**Course Code:** DSA0204 **Slot:** A

**Course Name:** COMPUTER VISION WITH OPENCV

**Course Faculty:** Dr.Senthil G A

**Project Title:** BridgeVisionNet: AI-Based Bridge Crack Detection and Structural Health Monitoring Using YOLOv12 and UNet++

**Module Photographs**:

| Architecture Diagram | Module3: Severity Classification & Structural Health Monitoring | Presentation Image |
|---|---|---|
|  |  |  |







**Project Description**:

**BridgeVisionNet: AI-Based Bridge Crack Detection and Structural Health Monitoring Using YOLOv12 and U-Net++** focuses on detecting, segmenting, and assessing cracks in bridge structures using computer vision and deep learning techniques. The system uses **YOLOv12** to identify crack regions and **U-Net++** to perform precise pixel-level crack segmentation.

**Module 3: Severity Classification & Structural Health Monitoring** analyzes the detected and segmented cracks using features such as crack length, width, area, and texture to classify their severity into **Low, Medium, High, and Critical** levels. The module provides structural health indicators, crack growth trends, risk assessment, visualization dashboards, and alerts for serious defects. This helps support early identification of structural problems and enables systematic monitoring and maintenance of bridge infrastructure.

**Student Signature** **Guide Signature**